

MEMORIAL MEDICAL CENTER

DEPARTMENT OF PATHOLOGY & LABORATORY MEDICINE

1750 Harrison Boulevard • Boston, MA 02118

Phone: (617) 555-0187 • Fax: (617) 555-0188

PATIENT NAME:	Hickle, Brady	DATE OF BIRTH:	11/27/1914
MEDICAL RECORD #:	MRN-8847293	AGE/SEX:	81 Years / Male
ACCESSION #:	P96-07854	SPECIMEN RECEIVED:	07/25/1996
ORDERING PHYSICIAN:	Dr. Marcus T. Whitmore, MD	REPORT DATE:	07/29/1996
LOCATION:	Urology Clinic, Memorial Medical Center		

SURGICAL PATHOLOGY REPORT

CLINICAL HISTORY:

81-year-old male with elevated PSA (8.2 ng/mL) and abnormal digital rectal examination. History of hypertension, diabetes mellitus, coronary heart disease, and chronic kidney disease. Referred for diagnostic prostate biopsy.

SPECIMEN SUBMITTED:

Prostate, transrectal ultrasound-guided needle core biopsies (12 cores)

PROCEDURE:

TRUS-guided prostate biopsy, 12-core sampling technique

GROSS DESCRIPTION:

Received fresh in twelve separately labeled containers are twelve prostate needle core biopsies, each designated by anatomic site. The cores are gray-tan to pink-tan tissue fragments measuring 0.8 to 1.5 cm in length and 0.1 cm in diameter. All tissue is submitted for microscopic examination in appropriately labeled cassettes.

Specimen A:

Right apex lateral – one core, 1.4 cm

Specimen B:

Right apex medial – one core, 1.3 cm

Specimen C:

Right mid lateral – one core, 1.2 cm

Specimen D:

Right mid medial – one core, 1.5 cm

Specimen E:

Right base lateral – one core, 1.1 cm

Specimen F:

Right base medial – one core, 1.3 cm

Specimen G:

Left apex lateral – one core, 1.2 cm

Specimen H:

Left apex medial – one core, 0.9 cm

Specimen I:

Left mid lateral – one core, 1.4 cm

Specimen J:

Left mid medial – one core, 1.1 cm

Specimen K:

Left base lateral – one core, 1.3 cm

Specimen L:

Left base medial – one core, 0.8 cm

MICROSCOPIC DESCRIPTION:

Sections are examined with hematoxylin and eosin staining. High-molecular-weight cytokeratin (34betaE12) immunohistochemistry is performed on selected cores to assist in the evaluation of atypical glands.

Core Location	Findings	Gleason Score	% Involvement
A – Right apex lateral	Benign prostatic tissue with mild chronic inflammation	N/A	—
B – Right apex medial	Benign prostatic tissue with focal glandular atrophy	N/A	—
C – Right mid lateral	Adenocarcinoma, prostatic acinar type	3+4=7 (Grade Group 2)	20%
D – Right mid medial	Benign prostatic tissue with benign prostatic hyperplasia	N/A	—
E – Right base lateral	Benign prostatic tissue	N/A	—
F – Right base medial	Benign prostatic tissue with focal chronic inflammation	N/A	—
G – Left apex lateral	Benign prostatic tissue	N/A	—
H – Left apex medial	Benign prostatic tissue with focal stromal hyperplasia	N/A	—
I – Left mid lateral	Adenocarcinoma, prostatic acinar type	3+4=7 (Grade Group 2)	15%
J – Left mid medial	Adenocarcinoma, prostatic acinar type		10%

Core Location	Findings	Gleason Score	% Involvement
		3+4=7 (Grade Group 2)	
K – Left base lateral	Benign prostatic tissue with benign prostatic hyperplasia	N/A	—
L – Left base medial	Benign prostatic tissue	N/A	—

Detailed Microscopic Findings – Malignant Cores:

Core C (Right mid lateral): This core shows infiltrative prostatic adenocarcinoma characterized by crowded small to medium-sized acini with enlarged nuclei, prominent nucleoli, and loss of basal cell layer. The tumor shows predominantly Gleason pattern 3 (well-formed glands) with a minor component of Gleason pattern 4 (fused glands and cribriform architecture), comprising approximately 20% of the core length. Immunohistochemistry confirms absence of basal cells in neoplastic glands (34betaE12 negative).

Core I (Left mid lateral): Adenocarcinoma is present, exhibiting similar histologic features with Gleason pattern 3 predominance and focal pattern 4 (poorly formed glands with incomplete lumina). Tumor involves approximately 15% of the core. Nuclear atypia is moderate with visible nucleoli.

Core J (Left mid medial): Adenocarcinoma occupies approximately 10% of the core, demonstrating predominantly Gleason pattern 3 architecture with a minor component of pattern 4 (fused and cribriform glands). The neoplastic glands show loss of basal cell layer and nuclear enlargement with prominent nucleoli.

Additional Findings:

Perineural invasion: Not identified in the examined sections

Extraprostatic extension: Cannot be assessed in needle core biopsies; no definitive extension is seen

High-grade prostatic intraepithelial neoplasia (HGPIN): Not identified

Seminal vesicle tissue: Not present in submitted specimens

DIAGNOSIS:

PROSTATE, TRANSRECTAL ULTRASOUND-GUIDED NEEDLE CORE BIOPSIES (12 CORES):

— **ADENOCARCINOMA, PROSTATIC ACINAR TYPE (3 of 12 cores positive)**

Gleason Score: 3+4=7 (Grade Group 2)

Involved cores: Right mid lateral (C), Left mid lateral (I), Left mid medial (J)

Tumor involvement: Approximately 15% overall (range 10-20% per positive core)

Perineural invasion: Not identified

Extraprostatic extension: Cannot be assessed on needle core biopsies

— Benign prostatic tissue with chronic inflammation and benign prostatic hyperplasia in remaining cores

COMMENT:

The findings are diagnostic of prostatic adenocarcinoma with Gleason score 3+4=7 (Grade Group 2, WHO/ISUP 2014 classification). The tumor involves 3 of 12 cores (25% of cores), predominantly in the mid-prostate regions bilaterally with approximate overall tumor burden of 15%. Gleason pattern 4 comprises a minor component in all positive cores.

Grade Group 2 prostatic adenocarcinoma is considered intermediate risk disease. The absence of perineural invasion in the examined tissue is noted, although this finding does not exclude its presence elsewhere in the prostate.

Extraprostatic extension cannot be reliably assessed in needle core biopsy specimens.

Clinical correlation with serum PSA levels, digital rectal examination findings, and radiologic imaging is recommended for treatment planning. Consideration for radical prostatectomy, radiation therapy, or active surveillance protocols should be discussed in the context of the patient's overall medical condition, comorbidities, and life expectancy.

The diagnosis was discussed with Dr. Whitmore on 07/29/1996.

Electronically signed by:

Dr. Patricia R. Steinberg, MD, PhD

Anatomic and Clinical Pathology

Board Certified in Anatomic Pathology and Urologic Pathology

Memorial Medical Center Department of Pathology

License #: MA-087653

Date of Report: July 29, 1996, 16:45 EDT

This is a final report. If you have any questions regarding this pathology report, please contact the Department of Pathology at (617) 555-0187.

*** **END OF REPORT** ***